Ex. no: 01

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Study of various Network commands used in linux and windows

Aim:

To study and practice various networking commands used in linux and windows for networking configuration, troubleshooting and diagnosis.

Commands:

1> ARP (Address Resolution Protocal) (arp -a):

Display and manages the ARP is short form of address resolution protocal, it will show the ip address of your computer along with your the ip address.

```
C:\Windows\system32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Windows\system32>
```

2) hostname: This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output : Ananthi

3). ipconfig: This command displays detailed configuration information about your TCP/IP connection including router, Gateway, DNS, DHCP and type of ethernet adapter in your system

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Npcap Loopback Adapter:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b923:c6f7:7122:8700%6
   Autoconfiguration IPv4 Address. . : 169.254.158.128
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::390c:dc1c:3ca7:94bc%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.10.86
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Windows\system32>_
```

4) nbtstat -a: This command helps solve problems with ~~metrics~~ netbios name resolution (Nbt stands for Net Bios over TCP/IP)

```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.


C:\Windows\system32>
```

5) netstat - (Network Statistics) netstat displays a statistics about a computer active TCP/IP connections. it is a command line tool for monitering network connections both incoming & outgoing as well as viewing routing tables, interfaces statistics etc.

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        DESKTOP-G0LVE4J:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-G0LVE4J:49675  ESTABLISHED
  TCP    127.0.0.1:49677        DESKTOP-G0LVE4J:49678  ESTABLISHED
  TCP    127.0.0.1:49678        DESKTOP-G0LVE4J:49677  ESTABLISHED
  TCP    172.16.10.86:13133     4.213.25.242:https     ESTABLISHED
  TCP    172.16.10.86:13199     lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.10.86:13258     a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.10.86:13259     150.171.110.82:https   CLOSE_WAIT
  TCP    172.16.10.86:13260     131.253.33.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13261     135.233.41.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13262     204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13270     72.154.187.226:https   CLOSE_WAIT
  TCP    172.16.10.86:13272     20.100.121.19:https    CLOSE_WAIT
  TCP    172.16.10.86:13273     a-0003:http            CLOSE_WAIT
  TCP    172.16.10.86:13278     13.107.42.254:https    CLOSE_WAIT
  TCP    172.16.10.86:13279     52.108.9.254:https     CLOSE_WAIT
  TCP    172.16.10.86:13281     172.202.65.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13293     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13294     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13295     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13297     bkk02s04-in-f14:https  ESTABLISHED
  TCP    172.16.10.86:13298     4.213.25.242:https     ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:1521  DESKTOP-G0LVE4J:49679  ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:49679  DESKTOP-G0LVE4J:1521   ESTABLISHED

C:\Windows\system32>_
```

6). nslookup : (name server lookup) is a tool used to perform DNS lookups in linux. It is used to displaying DNS details, such as the ip address of a particular computer, the MX records for a domain or the NS servers of a domain

```
C:\Windows\system32>nslookup www.google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4827:7700::
          2001:4860:482b:7700::
          2001:4860:4828:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4829:7700::
          2001:4860:482d:7700::
          2001:4860:482c:7700::
          142.251.156.119
          142.251.155.119
          142.251.151.119
          142.251.150.119
          142.251.157.119
          142.251.152.119
          142.251.154.119
          142.251.153.119

C:\Windows\system32>
```

7) Pathping - pathping is unique to windows, and is basically a combination of the ping and tracert commands. pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

```
C:\Windows\system32>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.

C:\Windows\system32>_
```

8) Ping - (Packet internet Groper) command is the best way to test connectivity between two nodes. ping use ICMP (Internet Control Message Protocol) to communicate to other devices.

1. # Ping hostname (Ping local host)
2. # Ping IP address (Ping 4.2.2.2)
3. # Ping fully qualified domain name.

```
C:\Windows\system32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t                Ping the specified host until stopped.
                      To see statistics and continue - type Control-Break;
                      To stop - type Control-C.
    -a                Resolve addresses to hostnames.
    -n count          Number of echo requests to send.
    -l size           Send buffer size.
    -f                Set Don't Fragment flag in packet (IPv4-only).
    -i TTL            Time To Live.
    -v TOS            Type Of Service (IPv4-only. This setting has been deprecated
                      and has no effect on the type of service field in the IP
                      Header).
    -r count          Record route for count hops (IPv4-only).
    -s count          Timestamp for count hops (IPv4-only).
    -j host-list      Loose source route along host-list (IPv4-only).
    -k host-list      Strict source route along host-list (IPv4-only).
    -w timeout        Timeout in milliseconds to wait for each reply.
    -R                Use routing header to test reverse route also (IPv6-only).
                      Per RFC 5095 the use of this routing header has been
                      deprecated. Some systems may drop echo requests if
```

94 Route :

Route command is used to show / manipulate the ip routing table. It is primarily used to setup static routes to specific host or networks via an interface

```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.
```

```
-c compartment Routing compartment identifier.
-p             Ping a Hyper-V Network Virtualization provider address.
-4             Force using IPv4.
-6             Force using IPv6.
```

```
If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*            .... Only prints those matching 157*

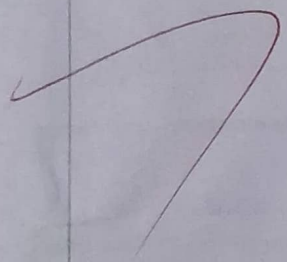
    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
              destination^      ^mask      ^gateway     metric^    ^
                                                        Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32

C:\Windows\system32>_
```



Linux Networking Commands.

1) ip

Display ip address

configure network interface

add / delete ip address

manage routing tables.

i) Display ip Address:

ip : Display all network interface and their assigned ip address

ii) ip address show:

Display the status of all network interface

iii) ip route:

Display the routing table and default gateway.

iv) 2. ip address add:

Assign a new ip address to a network interface.

```
[root@localhost ~]# ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                   macsec | maddress | monitor | mptcp | mroute | mrule |
                   neighbor | neighbour | netconf | netns | nexthop | ntable |
                   ntbl | route | rule | sr | stats | tap | tcpmetrics |
                   token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                    -h[uman-readable] | -iec | -j[son] | -p[retty] |
                    -f[amily] { inet | inet6 | mpls | bridge | link } |
                    -4 | -6 | -M | -B | -0 |
                    -l[oops] { maximum-addr-flush-attempts } | -echo | -br[ief] |
                    -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                    -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                    -c[olor]}
[root@localhost ~]#
```

```
[root@localhost ~]# ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:cb:73:6a brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83740sec preferred_lft 83740sec
    inet6 fe80::a00:27ff:fecb:736a/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[root@localhost ~]#
```

```
[root@localhost ~]# ip address add 192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip address del 192.168.1.254/24 dev enp0s3
[root@localhost ~]# ip link set enp0s3 up
[root@localhost ~]# ip link set enp0s3 down
[root@localhost ~]# ip link set enp0s3 promisc on

[deepak3@localhost ~]$ ip route get 10.10.1.4
10.10.1.4 via 10.0.2.2 dev enp0s3 src 10.0.2.15 uid 1000
    cache
```

3. Delete an ip Address:

Remove the ip address from the interface

Eg: ip address del 192.168.1.254/24 dev emp0s3

4. ip config:

It is staple in many sysadmin's tool belt for configuration configuring troubleshooting networks

5. mtr

combines the functionality of the ping and trace route commands.

a) mtr google.com:

shows you the statics, including each hop with time & loss %.

b). mtr -b google.com:

shows numeric ip address.

c) mtr -c 10 google.com:

set the number of ping that you want.

6. tcpdump:

this command is designed for capturing &

```
[root@localhost ~]# ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 106  bytes 7915 (7.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 106  bytes 7915 (7.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
                                 My traceroute  [v0.94]
localhost.localdomain (10.0.2.15) -> google.com                2026-07-24T23:00:18+0530
Keys:  Help   Display mode   Restart statistics   Order of fields   quit
                                                 Packets               Pings
 Host                                         Loss%   Snt   Last   Avg  Best  Wrst StDev
 1. _gateway                                  0.0%    18    1.8   1.7   0.9   3.5   0.7
 2. (waiting for reply)
 3. (waiting for reply)
 4. (waiting for reply)
 5. (waiting for reply)
 6. (waiting for reply)
 7. (waiting for reply)
 8. (waiting for reply)
 9. (waiting for reply)
10. pnmaaa-az-in-f14.1e100.net                0.0%    17   28.5  31.4  22.0  40.7   5.2
```

```
[root@localhost ~]# tcpdump
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL (Linux cooked v1), snapshot length 262144 bytes

^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
[root@localhost ~]# su -
[root@localhost ~]# dnf install -y tcpdump
CentOS Stream 9 - BaseOS                        0.0  B/s |   0  B     00:00
Errors during downloading metadata for repository 'baseos':
  - Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?rep
o=centos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.
centos.org]
Error: Failed to download metadata for repo 'baseos': Cannot prepare internal mirrorlist:
Curl error (6): Couldn't resolve host name for https://mirrors.centos.org/metalink?repo=ce
ntos-baseos-9-stream&arch=x86_64&protocol=https,http [Could not resolve host: mirrors.cent
os.org]
[root@localhost ~]#
```

i) tcpdump -i echo -c 10 host 8.8.8.8

Filter our traffic comming from a specific host.

7. ping ANANTHI :

It display best way to display test connecti-ivity b/w two nodes.

syntax : ping hostname.

Result :

The basic networking & linux commands are learned and executed successfully.

```
[root@localhost ~]# tcpdump -i enp0s3 -c 10 'host 8.8.8.8'
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
[root@localhost ~]# tcpdump -i enp0s3 net 10.1.0.0/24
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes
-v[v]
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
```

Output :

pinging ANANTHI [fe80::521f:7ba0:f461:]

with 32 bytes of data.

Reply from fe80::5218:7ba0:f461:cb3%.

: time < ms ping statics for fe80::521f::

7ba0:f461:cb3%.2:

packets : sent = 4, received = 4, lost = 0

minimum = 0ms Maximum = 0ms, Average = 0ms